

Hybrid CA for Avascular Tumour Growth: From Eco–Evolutionary Selection (Gerlee & Anderson, 2007) to Nutrient-Gradient-Driven Invasion (Messina et al., 2023)

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Introduction

Why these two papers

Both papers study **avascular tumour growth** with the same modelling backbone:

- Cells: a **cellular automaton (CA)** on a lattice;
- Nutrients: **continuous concentration fields** (e.g., oxygen/glucose);
- Transport: nutrient fields evolve via **reaction–diffusion PDEs**;
- Bidirectional coupling: nutrients *drive* cell actions; cells *consume/produce* nutrients.

They differ in the biological mechanism emphasized:

- **Gerlee & Anderson (2007)**: oxygen limitation drives **eco–evolutionary selection** (heritable variation in cell response).
- **Messina et al. (2023)**: glucose gradients drive **directional invasion/chemotaxis** and separate drivers of **growth vs invasiveness**.

Take-home: same hybrid CA/PDE scaffold $\Rightarrow$ clean comparison of *selection under oxygen scarcity vs migration under glucose gradients*.

Common mathematical scaffold (hybrid CA + PDE)

Lattice. $\Lambda = \{x_i\}_{i=1}^N \subset \Omega \subset \mathbb{R}^d$ (typically $d = 2$). One cell per site.

CA state. For each site x_i ,

$$s_i(t) \in \mathcal{S} \quad (\text{e.g., proliferative/quiescent/apoptotic/motile}).$$

Local update (deterministic or stochastic):

$$s_i(t + \Delta t) = \mathcal{U}\left(s_i(t), \{s_j(t)\}_{j \in \mathcal{N}(i)}, \mathbf{c}(x_i, t); \boldsymbol{\theta}\right).$$

Microenvironment fields. $\mathbf{c}(x, t) \in \mathbb{R}^m$ (oxygen, glucose, etc.) evolve by

$$\partial_t \mathbf{c} = \mathbf{D} \Delta \mathbf{c} + \mathbf{R}(\mathbf{c}, \{s_i\}),$$

where $\mathbf{R}$ encodes cell-dependent consumption/production.

Coupling. s_i depends on $\mathbf{c}(x_i, t)$; $\mathbf{R}$ depends on occupancy/state at sites.

Gerlee & Anderson (2007): oxygen-driven selection in a hybrid CA

Aim. Explain how oxygen availability shapes (i) tumour **morphology** and (ii) **clonal/phenotypic evolution** in avascular growth.

Key modelling choice: cell response network. Each cell carries a feed-forward map from local microenvironment to action:

$$\mathbf{y}_i(t) = F(\mathbf{u}_i(t); \mathbf{w}_i), \quad \mathbf{u}_i(t) = \mathbf{u}(\mathbf{c}(x_i, t)),$$

where $\mathbf{w}_i$ are cell-specific parameters (weights/thresholds).

Heritability + mutation. Upon division:

$$\mathbf{w}_{\text{child}} = \mathbf{w}_{\text{parent}} + \boldsymbol{\eta} \quad (\text{small random perturbation}).$$

Resource competition $\Rightarrow$ selection over $\mathbf{w}$ (eco–evolutionary dynamics).

Reported outcomes (qualitative). Oxygen levels shift tumour **shape** (compact vs fingered) and modulate emergent **aggressiveness/diversity**.

Messina et al. (2023): glucose gradients, chemotaxis, growth vs invasiveness

Aim. Quantify how **glucose gradients** affect spheroid **growth** and **invasiveness**, and identify dominant parameters for each.

Microenvironment: glucose field. A representative reaction–diffusion form:

$$\partial_t g = D_g \Delta g - \rho(s(x, t)) g,$$

with boundary conditions inducing either:

- **isotropic supply** (uniform boundary g),
- **anisotropic supply** (different boundary values) $\Rightarrow$ **gradient**.

CA dynamics with chemotaxis. Updates depend on local g and (for movement) on ∇g :

$$s_i(t + \Delta t) = \mathcal{U}(s_i(t), \{s_j\}_{j \in \mathcal{N}(i)}, g(x_i, t), \nabla g(x_i, t); \theta).$$

Reported outcomes (qualitative). Growth is mainly controlled by proliferation/adhesion-type parameters, while invasiveness is primarily driven by chemotactic sensitivity.

Comparison lens (what the joint story will emphasize)

Same backbone: CA on lattice + nutrient PDEs + coupling.

Different “cell policy” mechanisms:

- (2007) policy is **adaptive/evolvable**: $F(\cdot; \mathbf{w})$ with $\mathbf{w}$ mutating under selection.
- (2023) policy is **rule-based with gradient sensing**: explicit chemotaxis via ∇g .

Microenvironment as driver:

scarcity (oxygen) $\Rightarrow$ selection over strategies vs gradient (glucose) $\Rightarrow$ directed invasion.

Next (later slides): model specifics (states, boundary conditions), metrics (morphology/diversity/invasiveness), and a like-for-like comparison table.

Gerlee & Anderson (2007)

Model state and spatial discretization

Lattice (CA layer). Tissue is a 2D grid $\Lambda = \{(i, j)\}_{i,j=0}^{N-1}$ with spacing d (one cell per site). Each site stores an automaton element:

$$S_{i,j}(t) \in \{\emptyset, \text{alive cell}, \text{necrotic cell}\}.$$

Cell attributes (per occupied site). Each living cell carries (minimally):

- age $a(t)$ (for proliferation check),
- phenotype/decision mechanism: network parameters $\mathbf{w}$ (weights/thresholds),
- (optional) flags: quiescent/motile/metabolic mode (in full model).

Microenvironment fields (PDE layer on same grid). Continuous concentrations sampled on the same lattice:

$$c_{i,j}(t) \text{ (oxygen)}, \quad g_{i,j}(t) \text{ (glucose)}, \quad h_{i,j}(t) \text{ (H}^+),$$

with constant Dirichlet boundary conditions representing surrounding vasculature.

Reaction–diffusion PDEs and coupling to CA

Field dynamics (reaction–diffusion). For $i \in \{c, g, h\}$:

$$\partial_t i(\mathbf{x}, t) = D_i \Delta i(\mathbf{x}, t) \pm f_i(\mathbf{x}, t).$$

Consumption/production is cell-resolved:

$$f_i(\mathbf{x}, t) = \begin{cases} 0, & \text{if site empty,} \\ r_i F(\mathbf{x}), & \text{if occupied,} \end{cases}$$

where r_i are base rates and $F(\mathbf{x})$ is a modulated metabolic factor.

Discretization. PDEs are discretized by standard 5-point finite differences on spacing d and time step Δt :

$$\Delta i \approx \frac{i_{i+1,j} + i_{i-1,j} + i_{i,j+1} + i_{i,j-1} - 4i_{i,j}}{d^2}.$$

Interpretation. Nutrients diffuse on the grid; each occupied lattice point contributes a localized sink/source term (via f_i).

Cell decision rule: response network + mutation

Cell “policy” as a feed-forward response network. At site (i, j) sample local inputs (e.g., neighbour count and local nutrients):

$$\mathbf{x}_{i,j}(t) = \mathbf{x}(S_{\mathcal{N}(i,j)}(t), c_{i,j}(t), g_{i,j}(t), h_{i,j}(t)).$$

Compute network response (abstractly):

$$\mathbf{y}_{i,j}(t) = F(\mathbf{x}_{i,j}(t); \mathbf{w}_{i,j}),$$

whose argmax (or thresholded outputs) selects action

$$A \in \{\text{Proliferate (P), Quiesce (Q), Apoptose (A)}\} \quad (\text{and optionally Move/Metabolism in full model}).$$

Heritability and mutation at division. When a cell divides, daughter inherits network parameters with perturbation:

$$\mathbf{w}_{\text{child}} = \mathbf{w}_{\text{parent}} + \boldsymbol{\eta}, \quad \boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}) \text{ on mutated entries.}$$

Simulation loop and data-structure choices

Core loop (per global time step).

- ① Solve PDEs on the $N \times N$ grid (finite differences; Dirichlet boundaries).
- ② Create a list of living cells and **update in random order**.
- ③ For each cell:
 - sample $\mathbf{x}$ from local environment; compute action A via network;
 - execute:
 - **P**: if age $\geq A_p$ and empty neighbour exists, place daughter in a neighbouring site; else no division;
 - **Q**: mark quiescent state;
 - **A**: remove cell (site becomes empty).
 - **Necrosis**: starvation death leaves an immobile necrotic occupant (site remains blocked).
 - (optional) **Movement**: if enabled and not divided, attempt random move to empty neighbour.

Implementation view (minimal). Two coupled arrays:

gridState[i, j] (empty/alive/necrotic) $C[i, j]$ (oxygen field)

(+ G, H in full model), and a per-cell struct storing ($a, \mathbf{w}$, flags).

Results: morphology vs oxygen (hypoxia)

Control parameter. Tissue background oxygen concentration c_0 (or equivalently base oxygen consumption).

Emergent morphologies (qualitative).

- **Low oxygen (hypoxic), e.g. $c_0 = 10$:** diffusion-limited growth $\Rightarrow$ **fingered / branched** tumour boundary (DLA-like).
- **Higher oxygen, e.g. $c_0 = 2.5$ and above:** growth becomes **more regular / round**, with weaker fingering.

Mechanistic interpretation in the model. Local oxygen drop across the boundary makes proliferation a “race” towards higher c :

division at boundary $\Rightarrow$ local sink $\Rightarrow$ neighbour disadvantage $\Rightarrow$ tip growth.

Onset of cell death.

- Hypoxic case: cell death appears early ($t \approx 10$, ~ 7 days).
- Higher oxygen: hypoxic core appears later ($t \approx 35$, ~ 23 days).

Results: growth dynamics and viable rim structure

Tumour composition over time. After cell death begins, the tumour organizes into:

- necrotic/dead core (resource-deprived),
- **thin rim** of viable cells near boundary where $c(\mathbf{x}, t)$ remains sufficient.

Growth curves (qualitative).

- Hypoxic case ($c_0 = 10$): after early death, **approximately linear** increase in total cell count; proliferating/quiescent counts become roughly steady (limited to boundary tips).
- Higher oxygen: tumour grows larger before core death; then continues with dead-core + viable-rim dynamics.

Interpretation: oxygen availability sets an effective “active perimeter” that limits net growth once diffusion becomes limiting.

Results: population diversity (Shannon index)

Metric. Population diversity tracked via Shannon index

$$H(t) = - \sum_k p_k(t) \log p_k(t),$$

with p_k the fraction of clone k .

Time evolution differs by oxygen level.

- **Low oxygen** ($c_0 = 10$): H rises rapidly to ≈ 0.9 around $t \approx 30$ (~ 20 days), then slowly decreases to ≈ 0.8 by $t \approx 65$ days.
- **Higher oxygen**: H increases more slowly; stays low (≈ 0.2) while a quiescent core persists ($t \lesssim 40$), then increases after death onset and approaches ≈ 0.8 by the end.

Model explanation. Fingered morphology $\Rightarrow$ spatially separated “tips” $\Rightarrow$ quasi-independent evolution $\Rightarrow$ higher early diversity; round growth $\Rightarrow$ stronger competition $\Rightarrow$ slower diversity increase.

Results: evolved phenotypes under hypoxia (aggressiveness)

Phenotype proxy. Average response probabilities of the evolved networks (Table-level summary):

$$\bar{r} = (\bar{r}_P, \bar{r}_Q, \bar{r}_A), \quad \text{plus consumption multiplier } \bar{\gamma}.$$

Reported shift (high vs low oxygen).

- Under **normoxic/high** c_0 conditions: dominant clones remain close to ancestral response (small changes).
- Under **hypoxia/low** c_0 (e.g. $c_0 = 10$): clones evolve toward

$$\bar{r}_P \uparrow, \quad \bar{r}_Q \downarrow, \quad \bar{r}_A \downarrow,$$

plus increased consumption / shorter effective cell cycle (more “aggressive” strategy).

Interpretation: harsher microenvironment $\Rightarrow$ stronger selection pressure $\Rightarrow$ survival of fast-cycling, low-apoptosis phenotypes.

Gerlee & Anderson (2007) - extended model

GA2007 — Extended model: added fields and network I/O

Motivation. Include *glycolytic phenotype* and *tumour acidity* by adding glucose and hydrogen ions.

Input vector (extended).

$$\mathbf{x}(\tilde{\mathbf{x}}, t) = (n(\tilde{\mathbf{x}}, t), c(\tilde{\mathbf{x}}, t), g(\tilde{\mathbf{x}}, t), h(\tilde{\mathbf{x}}, t)),$$

where n is neighbour count, c oxygen, g glucose, h hydrogen ion concentration.

Output layer (extended). Besides life-cycle nodes (Proliferation/Quiescence/Apoptosis), the network includes:

- **Metabolic pathway node:** selects aerobic vs anaerobic metabolism,
- **Movement node:** enables movement if sufficiently large.

Continuous layer. Add PDE fields for glucose $g(\mathbf{x}, t)$ and hydrogen ions $h(\mathbf{x}, t)$ to the oxygen field $c(\mathbf{x}, t)$.

GA2007 — Extended model: metabolism coupling (rates + modulation)

Metabolic switch. Network decides aerobic vs anaerobic (via sign/threshold of metabolic response).

Base rates (pathway-dependent).

- Aerobic: consumes oxygen and glucose with base rates (r_c, r_g^a) .
- Anaerobic: consumes *more* glucose and *no* oxygen; produces hydrogen ions with rate r_h .

Response–metabolism modulation. Define a target response T_r and a modulation function (paper Eq. (1)):

$$F(R) = \max(k(R - T_r) + 1, 0.25),$$

where R is the life-cycle response magnitude (typically $R = \max\{P, Q\}$).

Coupling idea. Local consumption/production terms are scaled by $F(R)$:

$$\text{rates} \mapsto F(R) \times (\text{base rates}) \Rightarrow \text{cell-specific metabolism.}$$

GA2007 — Simulation: what changes in the full (extended) system

Minimal subsystem used first. To simplify analysis, the paper initially removes:

- movement,
- anaerobic metabolism choice,
- glucose and hydrogen-ion fields,

and studies oxygen-driven growth/evolution only.

Full extended loop (conceptually). Compared to the minimal loop, each global step additionally requires:

- ① solve **three** coupled reaction–diffusion fields (c, g, h) on the grid;
- ② per-cell: evaluate network outputs for (i) life-cycle and (ii) metabolism and (iii) movement;
- ③ update local sink/source terms using pathway-dependent rates and modulation $F(R)$.

Extra state carried by each cell. Metabolic mode (aerobic/anaerobic) and (optionally) movement-eligibility flag.

GA2007 — Extended model: what is (and is not) shown in results

Scope of reported simulations. The main simulation results in the paper are obtained with the **minimal** oxygen-only subsystem (no movement, no (g, h) fields), to isolate the effect of oxygen background c_0 .

Role of the extended model in the paper. The extended (c, g, h) + metabolism/movement machinery is introduced to:

- represent **glycolytic shift** (aerobic $\leftrightarrow$ anaerobic) and its cost in glucose uptake,
- represent **acidification** of the microenvironment via H^+ production,
- allow future experiments where g and h also drive selection via their inclusion in $\mathbf{x}$.

Take-away for your talk. You can present the extended model as a principled generalisation of the same hybrid CA/PDE scaffold, even if the paper's key plots focus on the minimal subsystem.

GA2007 — Metabolism coupling: effect on evolutionary trade-offs (from discussion)

Key mechanism introduced by modulation $F(R)$. Life-cycle response magnitude R simultaneously impacts:

- **proliferation potential** (via network outputs),
- **metabolic rate** (via $F(R)$ scaling of consumption/production).

Implication reported/discussed for hypoxic regimes. Under strong oxygen limitation, selection favours:

higher proliferation potential / faster cell cycle even if it increases consumption,

because boundary competition makes rapid division a dominant advantage.

Interpretation. The model couples *fitness* to a trade-off:

faster cycling $\longleftrightarrow$ higher metabolic load,

so scarcity can indirectly shape both phenotype and resource usage through the same network.

Messina et al. (2023)

Domain, layers, and state variables

Two-layer lattice. Square domain $2\text{ mm} \times 2\text{ mm}$ discretized into 100×100 lattice points (LPs), $\Delta = 20\text{ }\mu\text{m}$.

Cellular layer (CA). Each LP stores occupancy and phenotype:

$$S_{x,y}(t) \in \{\emptyset, \text{active}, \text{starved}, \text{necrotic}\},$$

plus an **adhesion state** (attached vs detached) for motility rules.

Glucose layer (PDE field). Glucose concentration $C(x, t)$ sampled on each LP:

$$C_{x,y}(t) \equiv C(x_{x,y}, t).$$

Initial/boundary conditions.

- isotropic: $C(\cdot, 0) = 5.5\text{ mM}$ and boundary fixed at 5.5 mM ;
- gradient: boundary fixed at 8 mM (left) and 3 mM (right), with initial linear profile (mean 5.5 mM).

Initial spheroid: radius $\approx 50\text{ }\mu\text{m}$ (21 cells) at domain center; simulated time $\approx 48\text{ h}$.

Glucose PDE, numerical method, and uptake term

Reaction–diffusion.

$$\partial_t C = D \Delta C - k(x, t),$$

with disparate time scales (diffusion seconds vs division hours) $\Rightarrow$ **adiabatic approximation**:

$$0 = D \Delta C - k(x, t) \quad (\text{pseudo-stationary}).$$

Finite differences on LP grid. Discretize Laplacian with 5-point stencil on spacing Δ .

Solver. Pseudo-stationary field solved each CA update using **SOR (simultaneous over-relaxation)** with **Chebyshev acceleration**.

Consumption (Michaelis–Menten, cell-localized).

$$k(x, t) = I(x, t) \frac{k_{\max} C(x, t)}{C(x, t) + K_m},$$

where $I(x, t) \in \{0, 1\}$ indicates LPs containing an *active* or *starved* cell (implemented via indicator/Dirac-delta representation on the grid).

CA rules: phenotype switching, migration, proliferation, death

Phenotype thresholds. Two glucose thresholds $C_{\text{starv}} > C_{\text{nec}}$:

$$\begin{cases} \text{active} & C_{x,y} > C_{\text{starv}}, \\ \text{starved} & C_{\text{nec}} < C_{x,y} < C_{\text{starv}}, \\ \text{necrotic} & C_{x,y} < C_{\text{nec}} \quad (\text{irreversible}). \end{cases}$$

Event clocks.

- migration every T_m ,
- duplication every T_d ,
- apoptosis if starved longer than T_{apop} (LP becomes empty).

Migration (8-neighbourhood + biased random walk). A cell is either *attached* or *detached*. For an attached cell, the probability to detach is

$$P_{\text{adh}} = \frac{N_{\text{cell}}^{\text{att}}}{K + N_{\text{cell}}^{\text{att}}}, \quad P_{\text{adh}} = 0 \text{ if detached,}$$

where $N_{\text{cell}}^{\text{att}}$ is the number of attached neighbouring cells and K controls adhesion strength.

Migration direction choice and proliferation rules

Direction choice considers the 9 candidate positions (8 neighbours + stay). Let C_5 be glucose at the current site and C_i at candidate position i . Define the relative change

$$A_i = \frac{C_i - C_5}{C_5}, \quad S_i = e^{\alpha A_i}, \quad s_i = \frac{S_i}{\sum_{j=1}^9 S_j},$$

with chemotactic sensitivity α . The move direction is sampled according to $\{s_i\}_{i=1}^9$ (via partition of $[0, 1]$ into intervals).

Proliferation. At each T_d , an *active attached* cell places a daughter in a random empty neighbour; if none exists, a **pushing** rule shifts cells along a path toward the cluster edge to create space.

Simulation process and concrete data structures

Per-step workflow.

- ① Given current cellular layer, compute consumption mask $I(x, t)$.
- ② Solve pseudo-stationary glucose field $C_{x,y}$ (SOR + Chebyshev).
- ③ Update each cell using local $C_{x,y}$:
 - phenotype transitions (active/starved/necrotic),
 - migration attempts at multiples of T_m (adhesion + chemotaxis),
 - proliferation attempts at multiples of T_d (placement/pushing),
 - apoptosis if starved duration $> T_{\text{apop}}$.

Minimal implementation view.

- `cellState[100][100]`: enum {empty, active, starved, necrotic}
- `attachFlag[100][100]`: bool (attached/detached)
- `timerM[100][100]`, `timerD[100][100]`, `starvTime[100][100]`
- `C[100][100]`: glucose field (float), updated by SOR iterations
- Random-number stream for stochastic events (migration direction, adhesion, etc.)

Replicates (stochasticity). Same parameter set is simulated multiple times and averaged (paper uses $n = 40$ replicates after convergence analysis).

Results: isotropic vs gradient glucose fields

Two experimental in silico settings.

- **Isotropic glucose:** uniform boundary supply $\Rightarrow$ no directional cue.
- **Gradient glucose:** boundary values impose $\nabla C \neq 0$ across domain $\Rightarrow$ directional cue for chemotaxis.

Observed qualitative behaviours.

- In isotropic case: outward migration is not directionally biased; invasion patterns remain approximately symmetric.
- In gradient case: cells adapt migration and exhibit **anisotropic invasion** aligned with the glucose gradient (chemotaxis-driven).

Key message: external gradient is not required to grow, but it is a strong driver for *directed* invasion.

Results: measured outputs (growth vs invasiveness)

Growth observables. The paper tracks tumour growth via:

- total number of cells $N(t)$,
- number of cells adhered to the core $N_a(t)$ (core size/area proxy).

Invasiveness observable. Let $N_d(t)$ be the number of detached/migrated cells. Define an invasiveness ratio (as in the paper):

$$I(t) = \frac{N_d(t)}{N_a(t)}.$$

Qualitative interpretation.

- Large N_a with small N_d : compact growth (low invasion).
- Moderate N_a with large N_d : dissolving core / strong invasion.

This separation (core vs detached) is used throughout the sensitivity analysis to disentangle growth and invasion.

Results: global sensitivity analysis (parameter ranking)

Parameters varied.

α (chemotactic sensitivity), K (adhesion), T_d (doubling time), T_m (migration interval).

Global sensitivity (main findings).

- **Isotropic field:** effect of α is **negligible**.
- **Gradient field:** α becomes **fundamental** in governing invasion (direction and magnitude).
- T_d and K are **critical** for core evolution and detachment in both fields; T_m has a similar but weaker influence.

High-level conclusion.

growth $\approx f(T_d, K)$ whereas invasion under gradients $\approx f(\alpha)$.

Results: local sensitivity / regimes and “optimal adhesion”

Local sensitivity highlights.

- $\alpha = 0 \Rightarrow$ gradient insensitivity (no directed migration).
- Increasing $\alpha \Rightarrow$ stronger alignment with positive gradient (chemotactic response).

Role of adhesion K : regime transition. There is a threshold-like behaviour around $K \approx 0.5$:

- smaller K : migration flux can dissolve the core over time;
- larger K : adhesion promotes core integrity and supports proliferation-dominated growth.

Perturbation effects (qualitative).

- decreasing T_d (faster division) promotes rapid growth (larger N , larger N_a);
- changing T_m modulates invasion/growth rates (more frequent migration accelerates dynamics).

Reported “optimal” adhesion insight. An intermediate K yields many migrating cells while maintaining approximately constant core size (trade-off growth vs detachment).

Comparison

Model comparison (I): common backbone and state variables

Common backbone (hybrid CA/PDE).

lattice CA for cells + reaction–diffusion PDE for nutrients + bidirectional coupling

Discrete layer.

- both: 2D lattice; one cell per site; neighbourhood-based constraints.
- Gerlee–Anderson (2007): cell carries *internal parameters* $\mathbf{w}$ (response network).
- Messina (2023): cell carries *phenotype/attachment flags* + event timers (rule-based).

Continuous layer.

- both: nutrient field(s) on same grid; boundary conditions emulate external supply.
- GA2007: oxygen (and optionally glucose, H^+) as microenvironment drivers.
- M2023: glucose-only driver with pseudo-stationary solve per CA step.

Coupling. Cells read $c(x_i, t)$ (or $C_{x,y}$) to choose actions; occupancy generates local sinks/sources in PDEs.

Model comparison (II): decision mechanism and key control parameters

Cell decision mechanism.

- GA2007: *evolvable policy* $\mathbf{y} = F(\mathbf{x}; \mathbf{w})$, heritable with mutation

$$\mathbf{w}_{\text{child}} = \mathbf{w}_{\text{parent}} + \boldsymbol{\eta}.$$

- M2023: *rule-based policy* with explicit chemotaxis

$$S_i = e^{\alpha A_i}, \quad A_i = \frac{C_i - C_5}{C_5}, \quad s_i = \frac{S_i}{\sum_j S_j},$$

and adhesion-controlled detachment

$$P_{\text{adh}} = \frac{N_{\text{cell}}^{\text{att}}}{K + N_{\text{cell}}^{\text{att}}}.$$

Dominant control knobs.

- GA2007: oxygen availability (c_0) sets selection pressure and morphology.
- M2023: α (chemotaxis), K (adhesion), T_d (doubling) separate invasion vs growth regimes.

Interpretation. Both encode microenvironment $\rightarrow$ behaviour, but via **evolution** (GA2007) vs **explicit gradient sensing** (M2023).

Results comparison (I): common qualitative patterns

Shared qualitative outcomes.

- **Microenvironment shapes morphology:** scarcity/gradients induce non-trivial tumour shapes (non-spherical boundaries, protrusions).
- **Spatial self-organization:** viable/active regions concentrate where nutrient is higher (typically near boundary or along gradient).
- **Separation of “core” vs “front”:**
 - GA2007: viable proliferative rim/tips vs deprived core.
 - M2023: adhered core vs detached migrating population (tracked by $N_{\text{cell}}^{\text{core}}$, $N_{\text{cell}}^{\text{mig}}$).

Common modelling message. Coupled discrete–continuous dynamics is sufficient to generate **emergent macroscopic phenotypes** from local rules.

Results comparison (II): key differences in what “emerges” and what is measured

Main emergent phenomenon.

- GA2007: **eco–evolutionary selection** under hypoxia

$$H(t) = - \sum_k p_k \log p_k \quad (\text{clonal diversity}) \quad \text{and evolved response shifts.}$$

- M2023: **directed invasion** under glucose gradients

$$\phi = \frac{N_{\text{cell}}^{\text{core}}}{N_{\text{cell}}^{\text{mig}}} \quad (\downarrow \phi \Rightarrow \uparrow \text{ invasiveness}),$$

plus anisotropy counts (N_N, N_S, N_E, N_W) to quantify directional bias.

Parameter-to-outcome logic.

- GA2007: varying c_0 changes *selection pressure* $\Rightarrow$ phenotype distribution + morphology.
- M2023: varying (α, K, T_d, T_m) yields a *regime map* separating growth vs invasiveness (via sensitivity analysis).

Conclusions

Conclusions — Strengths and limitations (model-level)

Gerlee & Anderson (2007)

- **Strengths:** explicit eco–evolutionary mechanism: heritable, mutable cell policy $\mathbf{y} = F(\mathbf{x}; \mathbf{w})$ with $\mathbf{w}_{\text{child}} = \mathbf{w}_{\text{parent}} + \boldsymbol{\eta}$; supports clonal diversity tracking (e.g., $H(t)$).
- **Limitations:** policy is flexible but less interpretable / harder to calibrate; directed gradient-sensing (chemotaxis) is not the primary driver (needs explicit addition for invasion-in-gradient questions).

Messina et al. (2023)

- **Strengths:** interpretable rule-based CA with explicit chemotaxis in glucose gradients and operational invasion metrics (core vs migrating populations, ϕ); systematic sensitivity analysis links parameters to outcomes.
- **Limitations:** no endogenous clonal evolution (rules/parameters do not mutate); heterogeneity is mainly parametric, not evolutionary.

Proposed unification (I): unified hybrid CA/PDE with evolvable policies

Goal: retain the common hybrid CA/PDE scaffold, add (i) evolvable decision rules (GA2007) and (ii) explicit gradient-driven migration (M2023).

State.

$$S_i(t) \in \{\emptyset, \text{active}, \text{starved}, \text{necrotic}\}, \quad \text{flags: attached/detached}, \quad \mathbf{w}_i \text{ (cell policy parameters)}.$$

Microenvironment fields (example: oxygen + glucose).

$$\partial_t c = D_c \Delta c - \rho_c(S) c, \quad \partial_t g = D_g \Delta g - \rho_g(S) \frac{g}{g + K_m},$$

with boundary conditions setting (i) baseline oxygen level and (ii) optional glucose gradient.

Evolvable cell policy (GA2007-style).

$$\mathbf{x}_i(t) = [c_i, g_i, \nabla g_i, n_{\text{occ}}, n_{\text{att}}, \dots], \quad \mathbf{y}_i(t) = F(\mathbf{x}_i(t); \mathbf{w}_i),$$

where $\mathbf{y}_i$ selects $A \in \{\text{Prolif}, \text{Quiesce}, \text{Die}, \text{Move}\}$.

Inheritance. On division:

$$\mathbf{w}_{\text{child}} = \mathbf{w}_{\text{parent}} + \boldsymbol{\eta}.$$

Proposed unification (II): chemotactic move operator + measurable outputs

Chemotactic move operator (M2023-style). If $A = \text{Move}$, evaluate the 9 candidate positions (8 neighbours + stay). Let g_5 be glucose at the current site and g_k at candidate k :

$$A_k = \frac{g_k - g_5}{g_5}, \quad S_k = e^{\alpha_i A_k}, \quad p_k = \frac{S_k}{\sum_j S_j}.$$

Sample move direction with probabilities $\{p_k\}$, subject to occupancy constraints.

Adhesion-controlled detachment (M2023-style). For attached cells:

$$P_{\text{adh}} = \frac{N_{\text{cell}}^{\text{att}}}{K + N_{\text{cell}}^{\text{att}}}.$$

Detachment state affects motility and cluster integrity.

What can be tested/quantified in the unified model.

- **Evolution:** clonal diversity $H(t) = -\sum_k p_k \log p_k$, distribution of $\mathbf{w}$ (and derived traits such as α_i).
- **Invasion:** core vs migrating populations and invasiveness metric

$$\phi = \frac{N_{\text{cell}}^{\text{core}}}{N_{\text{cell}}^{\text{mig}}} \quad (\downarrow \phi \Rightarrow \uparrow \text{ invasion}).$$

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